磁场对电流元的作用力

$$d\vec F = I\, d\vec l \times \vec B$$

对于环电流线：

$$F = \oint_L d\vec F = I \oint_L d\vec l \times \vec B = I\left(\oint_L d\vec l\right)\times \vec B = 0$$

$$d\vec \tau = \vec r \times d\vec F = \vec r \times (I\, d\vec l \times \vec B)$$

$$\vec \tau = \oint_L d\vec \tau = I \oint_L \vec r \times (d\vec l \times \vec B)$$

$$= I \oint_L [d\vec l(\vec r\cdot\vec B) - \vec B(\vec r\cdot d\vec l)] \qquad d\vec l = d\vec r$$

$$= I \oint_L [(\vec r\cdot\vec B)\, d\vec l - \vec B\, d(r^2)]$$

$$= I \oint_L (\vec r\cdot\vec B)\, d\vec l = I \iint$$

$$= I \vec S \times \vec B = \vec m \times \vec B$$

$$\vec S = \oint_L dS = \oint_L \frac{1}{2}\vec r\cdot d\vec l\cdot \vec n = \frac{1}{2}\oint_L \vec r\times d\vec l$$

格林公式：

$$\oint_L \vec r\times d\vec l = \oint_L -y\,dx + x\,dy = \iint_D [1-(-1)]\,dx\,dy = 2S$$

对于连续电流线：

$$I = \vec J\cdot\vec S$$

$\vec J, \vec S, d\vec l$ 平行

$$\to I\, d\vec l = (\vec J\cdot\vec S)\, d\vec l = \vec J\, dV$$

$$\vec m = I\vec S$$

$$= \frac{1}{2} I \oint_L \vec r\times d\vec l$$

$$= \frac{1}{2}\oint_L \vec r\times \vec J\, dV$$

$$\Rightarrow d\vec m = \frac{1}{2}\vec r\times\vec J\, dV$$

B-S 定律

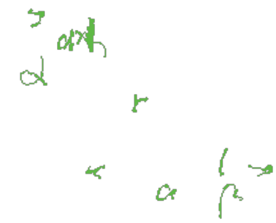
$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{l}\times\vec{r}}{r^3} = \frac{\mu_0}{4\pi}\frac{\vec{J}\,dV\times\vec{r}}{r^3}$$

对称性：

① $e\to -e$ $I\to -I$ $\vec{B}\to -\vec{B}$

② $\vec{r}\to -\vec{r}$ $\vec{B}\to -\vec{B}$ （$\vec{B}$是轴矢量）

例：求载流直导线的磁场



$$dB = \frac{\mu_0}{4\pi}\frac{I\,dx\sin\alpha}{r^2} = \frac{\mu_0 I}{4\pi}\frac{\frac{a\,d\alpha}{\sin^2\alpha}\sin\alpha}{\frac{a^2}{\sin^2\alpha}} = \frac{\mu_0 I}{4\pi a}\sin\alpha\,d\alpha$$

$$\vec{B} = \int_{\alpha_1}^{\alpha_2} dB$$

$$= \frac{\mu_0 I}{4\pi a}(\cos\alpha_1 - \cos\alpha_2) = \frac{\mu_0 I}{4\pi a}(\sin\beta_1 - \sin\beta_2)$$

推论：

无限长直导线：$\alpha_1 = 0$ $\alpha_2 = \pi$

$$B = \frac{\mu_0 I}{2\pi a}$$

半无限长直导线：$\alpha_1 = \frac{\pi}{2}$ $\alpha_2 = \pi$

$$B = \frac{\mu_0 I}{2\pi a}$$

例：载流圆线圈轴上的磁场

$$dB_{//} = \frac{\mu_0}{4\pi}\frac{I\,dl}{(r_0^2+x^2)^{3/2}}\cdot r_0$$

$$B = \frac{\mu_0 I r_0^2}{2(r_0^2+x^2)^{3/2}} = \frac{\mu_0 I r_0^2}{2r^3}$$

$$= \frac{\mu_0}{2\pi}\frac{m}{r^3}$$

例：密绕螺线管

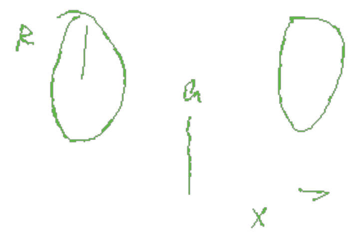
i 为单位长度电流

$$dB = \frac{\mu_0}{2}\frac{dI R^2}{\left(\frac{R}{\sin\alpha}\right)^3} = \frac{\mu_0 i}{2}\sin^3\alpha \, d\left(\frac{1}{\tan\alpha}\right)$$

无限： $B = \frac{\mu_0 i}{2}\int_0^{\pi}\sin\alpha \, d\alpha = \mu_0 i = \mu_0 n I$

半无限： $B = \frac{\mu_0 n I}{2}$

例：亥姆霍兹线圈.



$$B = \frac{\mu_0 I}{2}R^2 \cdot \frac{1}{\left[R^2+\left(\frac{a}{2}+x\right)^2\right]^{3/2}} + \frac{\mu_0 I}{2}R^2 \cdot \frac{1}{\left[R^2+\left(\frac{a}{2}-x\right)^2\right]^{3/2}}$$

$$\frac{dB}{dx} = \frac{\mu_0 I R^2}{2}\left\{\frac{-\frac{3}{2}\cdot 2\left(\frac{a}{2}+x\right)}{\left[R^2+\left(\frac{a}{2}+x\right)^2\right]^{5/2}} + \frac{\frac{3}{2}\cdot 2\left(\frac{a}{2}-x\right)}{\left[R^2+\left(\frac{a}{2}-x\right)^2\right]^{5/2}}\right\}$$

$$\left.\frac{dB}{dx}\right|_{x=0} = 0$$

$$\frac{d^2B}{dx^2} = -\frac{3\mu_0 I R^2}{2}\left[\frac{\left[R^2+\left(\frac{a}{2}+x\right)^2\right]^{5/2} - 5\left[R^2+\left(\frac{a}{2}+x\right)^2\right]^{3/2}\left(\frac{a}{2}+x\right)^2}{\left[R^2+\left(\frac{a}{2}+x\right)^2\right]^5} - \frac{-\left[R^2+\left(\frac{a}{2}-x\right)^2\right]^{5/2} + 5\left[R^2+\left(\frac{a}{2}-x\right)^2\right]^{3/2}\left(\frac{a}{2}-x\right)^2}{\left[R^2+\left(\frac{a}{2}-x\right)^2\right]^5}\right]$$

$$\left.\frac{d^2B}{dx^2}\right|_{x=0} = -\frac{3\mu_0 I R^2}{2}\left[\frac{R^2+\left(\frac{a}{2}\right)^2-5\left(\frac{a}{2}\right)^2}{\left[R^2+\left(\frac{a}{2}\right)^2\right]^{7/2}} + \frac{R^2+\left(\frac{a}{2}\right)^2-5\left(\frac{a}{2}\right)^2}{\left[R^2+\left(\frac{a}{2}\right)^2\right]^{7/2}}\right]$$

$$= -3\mu_0 I R^2 \frac{R^2-a^2}{\left[R^2+\left(\frac{a}{2}\right)^2\right]^{7/2}}$$

⇒ $R=a$ 时，磁场最均匀.

磁高斯定理

$$\nabla \cdot \vec{B} = 0$$

简略证明：磁场线总是闭合

$\Leftrightarrow$ ①穿入又穿出 ②未穿过 $\Rightarrow \sum \Phi = \sum \vec{B} \cdot d\vec{S} = 0 \Rightarrow \nabla \cdot \vec{B} = 0$

对于空间电流体：

$$\vec{B} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J}(\vec{r}') \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} d\tau'$$ 其中 $\vec{r}'$ 是源点坐标

$$\nabla \cdot \vec{B} = \frac{\mu_0}{4\pi} \int_{V'} \nabla \cdot \left(\frac{\vec{J}(\vec{r}') \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3} \right) d\tau'$$

$$= \frac{\mu_0}{4\pi} \int_{V'} \left[\nabla \times \vec{J}(\vec{r}') \cdot \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} - \vec{J}(\vec{r}') \cdot \left(\nabla \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right) \right] d\tau'$$

$\nabla \times \vec{J}(\vec{r}') = 0$ （∇ 针对的是 $\vec{r}$）

$$\nabla \times \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} = \nabla \times \nabla \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = 0$$

$\Rightarrow \nabla \cdot \vec{B} = 0$

安培环路定理，

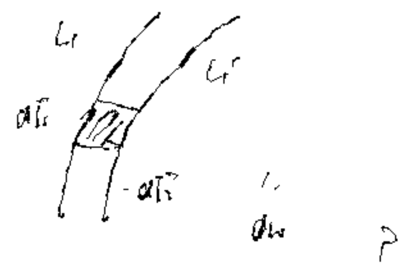
相信一定倚角

$$\vec B(\vec r) = \frac{\mu_0 I}{4\pi}\oint_{L_1}\frac{d\vec l_1\times\vec r_1}{r_1^3}$$

当P点移动 $d\vec r$ $\Leftrightarrow$ L_1移动 $-d\vec r$ 变为 L_1'

$\vec B(r)\cdot(-d\vec r)$

$$= \frac{\mu_0 I}{4\pi}\oint_{L_1}\frac{-d\vec r\cdot(d\vec l_1\times\vec r_1)}{r_1^3} = \frac{\mu_0 I}{4\pi}\oint_{L_1}\frac{(-d\vec r\times d\vec l_1)\cdot\vec r_1}{r_1^3}$$



$$= \frac{\mu_0 I}{4\pi}\oint_{L_1}\frac{d\vec S\cdot\vec r_1}{r_1^3} = -\frac{\mu_0 I}{4\pi}\oint_{L_1}\frac{d\vec S\cdot\vec r_{21}}{r_{21}^3}$$

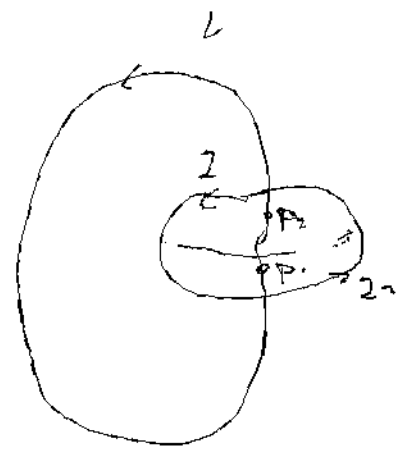
$$= -\frac{\mu_0 I}{4\pi}\oint_{L_1}d\omega = -\frac{\mu_0 I}{4\pi}\omega$$

记 S，S' 分别为 L_1，L_1' 的蒙面.

因P在外部

故 $\Omega_{S-P} - \Omega_{S'-P} + \omega = 0$

$\omega = \Omega_{S'-P} - \Omega_{S-P} = \underline{d\vec r\cdot\nabla\Omega}$ 不是 $-d\vec r\cdot\nabla\Omega$



$\Rightarrow \vec B = \frac{\mu_0 I}{4\pi}\nabla\Omega$

$$\oint_L\vec B\cdot d\vec l = \int_{P_1}^{P_2}\vec B\cdot d\vec l + \int_{P_2}^{P_1}\vec B\cdot d\vec l$$

$P_2\to P_1$ 时

$$\int_{P_1}^{P_2}\vec B\cdot d\vec l = 0$$

$$\int_{P_2}^{P_1}\vec B\cdot d\vec l = \frac{\mu_0 I}{4\pi}\int_{P_2}^{P_1}\nabla\Omega\cdot d\vec l = \frac{\mu_0 I}{4\pi}(\Omega_1-\Omega_2) = \mu_0 I$$

例：长直圆柱面.

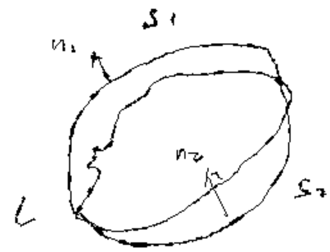
例：载流长直螺线管.

例：载流螺绕环

磁矢势 $\vec{A}$

$\nabla\cdot\vec{B}=0$ $\quad\longrightarrow\quad$ $\nabla\cdot(\nabla\times\vec{A})=0$

$\nabla\times\vec{B}=\mu_0\vec{J}$ $\quad$ 可以引入磁矢势



$$\oiint_{S_1+S_2}\vec{B}\cdot d\vec{S}=\iint_{S_1}\vec{B}\cdot d\vec{S_1}-\iint_{S_2}\vec{B}\cdot d\vec{S_2}=0$$

$\Rightarrow \iint_{S_1}\vec{B}\cdot d\vec{S}=\iint_{S_2}\vec{B}\cdot d\vec{S}$ 磁通量仅与S的边界L有关

定义 $\vec{B}=\nabla\times\vec{A}$

$$\iint_S\vec{B}\cdot d\vec{S}=\oint_L\vec{A}\cdot d\vec{l}$$

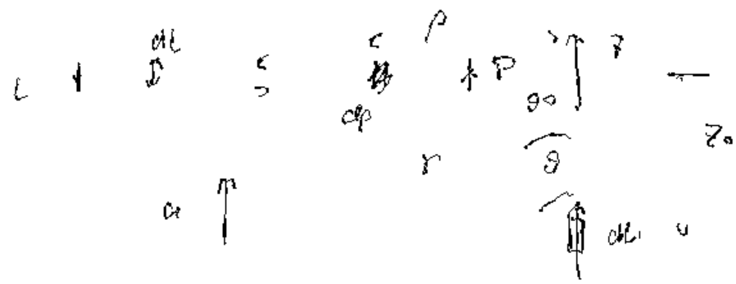
$\vec{A}$ 不唯一 (有冗余/规范对称性)

$\nabla\times\nabla\phi=0$

$\vec{A'}=\vec{A}+\nabla\phi$

$\vec{B}=\nabla\times\vec{A'}=\nabla\times\vec{A}$

一般取库仑规范势 $\nabla\cdot\vec{A}=0$



$$\oint_L\vec{A}\cdot d\vec{l}=A_z(P)\cdot dl$$

$$\phi_L=\iint_S\vec{B}\cdot d\vec{S}=\iint_S dB(r)\,d\rho\,dl$$

$$dB(r)=\frac{\mu_0 I}{4\pi}\frac{dl_1\sin\theta}{r^2}=\frac{\mu_0 I}{4\pi}\frac{\cos^2\theta}{z_0^2}\sin\theta\,dl_1$$

$$\phi_L=\oiint\frac{\mu_0 I}{4\pi z_0^2}\cdot\cos^2\theta\sin\theta\cdot z_0\frac{d\theta}{\cos^2\theta}\,dl\,dl_1=\int_{\theta_0}^{\frac{\pi}{2}}\frac{\mu_0 I}{4\pi z_0}\sin\theta\,d\theta\,dl\,dl_1$$

$$=\frac{\mu_0 I}{4\pi z_0}\cos\theta_0\,dl\,dl_1$$

$$\oint_L A\cdot d\vec{l}=dA(z)\cdot dl$$

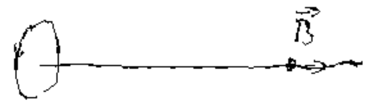
$$\Rightarrow dA(z)=\frac{\mu_0 I\cos\theta_0}{4\pi z_0}dl_1=\frac{\mu_0 I\,dl_1}{4\pi r_0}$$

$\vec{A}(r) = \frac{\mu_0 I}{4\pi}\oint \frac{d\vec{l}}{r}$

磁矩

$\vec{m} = I\vec{S}$

$\vec{B_r} = \frac{\mu_0 I}{2R} = \frac{\mu_0 m}{2\pi R^3}$

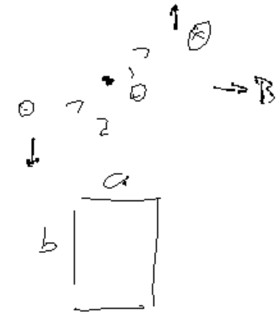


一般形式：

$\vec{B} = \frac{\mu_0}{4\pi}\left[-\frac{\vec{m}}{r^3} + \frac{3(\vec{m}\cdot\vec{r})\vec{r}}{r^5}\right]$ （磁偶极子～电偶极子）

磁力矩

$L = 2BIb\cdot\frac{a}{2}\cos\theta = BIS\cos\theta$



$\vec{L} = (I\cdot S\vec{n})\times\vec{B} = \vec{m}\times\vec{B}$

定义

磁势能

$\frac{\partial U}{\partial\theta} = L = \vec{m}\times\vec{B}$

$\Rightarrow U = U_0 + \int_{\theta_0}^{\theta}\vec{m}\times\vec{B}\,d\theta$

$= U_0 + mB\int_{\theta_0}^{\theta}\sin\theta\,d\theta = U_0 + mB\cos\theta_0 - mB\cos\theta$

$U_m := -\vec{m}\cdot\vec{B}$

则有 $F = -\nabla U_m = -\nabla(-\vec{m}\cdot\vec{B}) = \nabla(\vec{m}\cdot\vec{B})$

↳ 磁场将磁偶极子从 B 小处移到 B 大处

例：不均匀磁场中线圈受力问题.

$F = -$

$B_r\cdot 2\pi r\,dz + [B_z(z+dz) - B_z(z)]\pi r^2 = 0$

$\Rightarrow B_r = -\frac{r}{2}\frac{dB_z}{dz}$

$F = \pi r^2 I\frac{dB_z}{dz} = m\frac{dB_z}{dz}$

$(F = -\nabla(-\vec{m}\cdot\vec{B}) = m\frac{dB_z}{dz})$

例 5.1.4)

(1) 求轴上距圆心 L 处的磁感应强度 B

$$dB = \frac{\mu_0}{2\pi}\frac{dm}{(r^2+L^2)^{3/2}}$$

$$= \frac{\mu_0}{2\pi}\frac{\sigma\cdot 2\pi r\,dr}{(r^2+L^2)^{3/2}}\frac{\omega}{2\pi}\cdot\pi r^2 \qquad -\frac{1}{2}\frac{1}{(r^2+L^2)^{3/2}}$$

$$B = \int_0^R dB(r)$$

$$= \frac{\mu_0\sigma}{2}\omega\int_0^R\frac{r^3\,dr}{(r^2+L^2)^{3/2}}$$

$$\int_0^R\frac{r^3\,dr}{(r^2+L^2)^{3/2}} = L\int_0^{\frac{R}{L}}\frac{k^3\,dk}{(1+k^2)^{3/2}}$$ 令 $k=\frac{r}{L}$，$k=\tan t$

$$= L\int_0^{\frac{R}{L}}\tan^3 t\cdot\cos^3 t\cdot\frac{dt}{\cos^2 t} = L\int_0^{\frac{R}{L}}\frac{\sin^3 t}{\cos^2 t}\,dt$$

$$= L\cdot -\int_0^{\frac{R}{L}}\frac{(1-\cos^2 t)}{\cos^2 t}\,d(\cos t) = L\left(\frac{1}{\cos t}+\cos t\right)\Bigg|_{t=0}^{t=\arctan\frac{R}{L}}$$

$$= L\left(\frac{\sqrt{L^2+R^2}}{L}+\frac{L}{\sqrt{L^2+R^2}}-2\right) = \sqrt{L^2+R^2}+\frac{L^2}{\sqrt{L^2+R^2}}-2L$$

$$B = \frac{1}{2}\mu_0\sigma\omega\left(\sqrt{L^2+R^2}+\frac{L^2}{\sqrt{L^2+R^2}}-2L\right)$$ 与ppt不同

(2) 求磁矩 m

$$m = \int_0^R\sigma\cdot 2\pi r\,dr\cdot\frac{\omega}{2\pi}\cdot\pi r^2 = \frac{1}{4}\pi\sigma\omega R^4$$

03-19

例 6.1.4 均匀带电量为 Q，半径为 R 的球绕轴以 ω 旋转，

求球内外 B 和磁矩 m

$$\sigma = \frac{Q}{4\pi R^2}$$

$$dB = \frac{\mu_0}{2\pi}\frac{dm(\theta)}{r^3}$$

$$= \frac{\mu_0}{2\pi}\frac{\pi\sigma\omega R^4 \sin^3\theta\, d\theta}{(R^2+x^2+2Rx\cos\theta)^{3/2}}$$

$$= \frac{1}{2}\mu_0\sigma\omega R^4 \frac{\sin^3\theta\, d\theta}{(R^2+x^2+2Rx\cos\theta)^{3/2}}$$

$$\int \frac{dx}{\sqrt{a+bx}} = \frac{2\sqrt{a+bx}}{b}$$

$$\int \frac{x\,dx}{\sqrt{a+bx}} = -\frac{2(2a-3b)}{3b^2}\sqrt{a+bx}$$

$$\int \frac{x^2\,dx}{\sqrt{a+bx}} = \frac{2(8a^2-4abx-3b^2x^2)}{15b^3}\sqrt{a+bx}$$

$a = (R+x)^2$, $b = 2Rx$

$k = \cos\theta - 1$

$$B = \int dB$$

$$= \frac{1}{2}\mu_0\sigma\omega R^4 \int_0^{\pi} \frac{(\cos\theta-1)(\cos\theta+1)\,d(\cos\theta-1)}{[(R+x)^2+2Rx(\cos\theta-1)]^{3/2}}$$

$k = \cos\theta - 1$

$$= \frac{1}{2}\mu_0\sigma\omega R^4 \int_0^{-2} \frac{(k^2+2k)\,dk}{(\sqrt{a+bk})^3}$$

$$\int_0^{-2} \frac{(k^2+2k)\,dk}{(a+bk)^{3/2}}$$

$$= \int_0^{-2} \frac{(k^2+2k)\,dk}{(a+bk)} \cdot \frac{1}{\sqrt{a+bk}}$$

$$= \int_0^{-2} \frac{1}{a+bk}\left[k^2+\frac{a}{b}k+\left(2-\frac{a}{b}\right)\left(k+\frac{a}{b}\right)-\left(2-\frac{a}{b}\right)\frac{a}{b}\right]\cdot\frac{dk}{\sqrt{a+bk}}$$

$$= \int_0^{-2}\left(\frac{1}{b}k+\frac{1}{b}\left(2-\frac{a}{b}\right)-\frac{\frac{a}{b}\left(2-\frac{a}{b}\right)}{a+bk}\right)\frac{dk}{\sqrt{a+bk}}$$

$$= \frac{1}{b}\int_0^{-2}\frac{k\,dk}{\sqrt{a+bk}} + \frac{1}{b}\left(2-\frac{a}{b}\right)\int_0^{-2}\frac{dk}{\sqrt{a+bk}} - \frac{a(2b-a)}{b^2}\int_0^{-2}\frac{dk}{(a+bk)^{3/2}}$$

$$= \frac{1}{b}\left\{\left[-\frac{2(2a-bk)}{3b^2}\sqrt{a+bk}\right]_0^{-2} + \left[\frac{2b-a}{b^2}\cdot\frac{2}{b}\sqrt{a+bk}\right]_0^{-2} - \left[\frac{a(2b-a)}{b^3}(-2)\frac{1}{\sqrt{a+bk}}\right]_0^{-2}\right.$$

$$= \frac{2}{3b^3}\left[2a\sqrt{a} - (2a+2b)\sqrt{a-2b}\right] + \frac{2(2b-a)}{b^3}\left(\sqrt{a-2b}-\sqrt{a}\right) + \frac{2a(2b-a)}{b^3}\left(\frac{1}{\sqrt{a-2b}} - \frac{1}{\sqrt{a}}\right)$$

$$= \frac{1}{b^3}\Big\{ \frac{4}{3}a^{3/2} - \frac{4}{3}(a+b)\sqrt{a-2b} + 4b\sqrt{a-2b} - 2a\sqrt{a-2b} - 4b\sqrt{a} + 2a^{3/2}$$

$$- 2a\sqrt{a-2b} - 4b\sqrt{a} + 2a^{3/2} \Big]$$

$$= \frac{1}{b^3}\left[\frac{16}{3}a^{3/2} - \frac{16}{3}a\sqrt{a-2b} + \frac{8}{3}b\sqrt{a-2b} - 8b\sqrt{a}\right] \quad a = (R+x)^2,\ b = 2Rx$$

① $x<R$

$$= \frac{1}{8R^3x^3}\left[\frac{16}{3}(R+x)^3 - \frac{16}{3}(R+x)^2(R-x) + \frac{16}{3}Rx(R-x) - 16Rx(R+x)\right]$$

$$= \frac{1}{8R^3x^3}\left[\frac{32}{3}(R+x)^2 x - \frac{32}{3}Rx(R+2x)\right]$$

$$= \frac{4}{3R^3x^2}\left[(R+x)^2 - R(R+2x)\right] = \frac{4}{3R^3x^2}x^2$$

$$= \frac{4}{3R^3}$$

$$B = \frac{1}{2}\mu_0\sigma\omega R^4 \int_0^{-2}\frac{(k^2+2k)\,dk}{(\sqrt{a+bk})^3}$$

$$= \frac{1}{2}\mu_0\sigma\omega R^4 \frac{4}{3R^3} = \frac{2}{3}\mu_0\sigma\omega R = \frac{\mu_0 Q\omega}{6\pi R} \quad (x<R)$$

② $x>R$

$$\frac{1}{b^3}\left[\frac{16}{3}a^{3/2} - \frac{16}{3}a\sqrt{a-2b} + \frac{8}{3}b\sqrt{a-2b} - 8b\sqrt{a}\right]$$

$$= \frac{1}{8R^3x^3}\left[\frac{16}{3}(R+x)^3 - \frac{16}{3}(R+x)^2(x-R) + \frac{16}{3}Rx(x-R) - 16Rx(R+x)\right]$$

$$= \frac{1}{8R^3x^3}\left[\frac{32}{3}(R+x)^2 R - \frac{32}{3}Rx(x+2R)\right] = \frac{4}{3R^2x^3}\left[(R+x)^2 - x(x+2R)\right]$$

$$= \frac{4}{3x^3}$$

$$B = \frac{\mu_0 Q\omega}{6\pi R}\cdot\frac{R^3}{x^3} = \frac{\mu_0 Q\omega R^2}{6\pi x^3} \quad (x>R)$$

$$dm(\theta) = \sigma\cdot 2\pi R^2\sin\theta\,d\theta\cdot\frac{\omega}{2\pi}\cdot\pi R^2\sin^2\theta$$

$$= \pi\sigma\omega R^4\sin^3\theta\,d\theta$$

$$m = \int_0^{\pi} dm(\theta) = \pi\sigma\omega R^4\int_0^{\pi}\sin^3\theta\,d\theta$$

$= \frac{1}{4} Q \omega R^4 \int_0^{\pi} [\cos^2\theta - 1] \, d(\cos\theta)$

$= \frac{1}{4} Q \omega R^4 \quad \frac{1}{3}\cos^3\theta - \cos\theta \Big|_{\cos\theta = 1}^{\cos\theta = -1}$

$= \frac{1}{4} Q \omega R^4 \cdot \frac{4}{3} = \frac{1}{3} Q \omega R^4$

05-1

例 5.1.4

电荷量 Q 的匀地分布在半径为 R 的球体内，球体以角速度 ω 匀速绕直径旋转、求（1）距离球心 r 处的电流密度 j

$\rho = \frac{3Q}{4\pi R^3}$

$r \sim r + dr$ 球壳内

$dI = j \cdot dS = j \cdot 2\pi r \, dr$

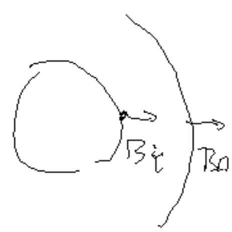
$= \frac{dQ}{T} = \frac{\omega}{2\pi} \rho \cdot 4\pi r^2 dr$

得 $j = \frac{\omega}{2\pi} \cdot 2\rho r = \frac{\omega\rho}{\pi} r = \frac{3Q\omega}{4\pi^2 R^3} r$

$* \ \vec{j} = \rho\vec{v} = \rho\,\vec{\omega} \times \vec{r}$?

(2)

由 5.1.4 知，



对于球内，$r < R$

$B(r) = \int dB$

$= \int_0^r dB + \int_r^R dB$

$= \int_0^r \frac{\mu_0 \omega r'^2}{6\pi r^3} dQ + \int_r^R \frac{\mu_0 \omega \, dQ}{6\pi r'^3}$

$= \frac{\mu_0 \omega \rho}{6\pi} \int_0^r \frac{1}{r^3} 4\pi r'^4 dr' + \int_r^R \frac{1}{r'} 4\pi r'^2 dr'$

$= \frac{2}{3} \mu_0 \omega \rho \left[\frac{1}{5} r^2 + \frac{1}{2}(R^2 - r^2) \right]$

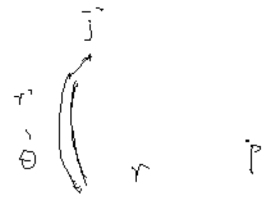
$= \frac{\mu_0 \omega Q}{20\pi R^3} (5R^2 - 3r^2)$ $\qquad 2r^2$

对于球外，$r > R$

$$B = \int dB$$

$$= \int_0^R \frac{\mu_0 \omega r'^2}{6\pi r^3} \rho \cdot 4\pi r'^2 \, dr' = \frac{\mu_0 \omega Q}{10\pi r^2} R^2$$

例 5.1.50



$$\vec{J}(r) = \frac{1}{4\pi r^2} 2eN \hat{r}' = \frac{eN}{2\pi r^2} \hat{r}'$$

助教打错了!

$$dB(r') = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

$I d\vec{l} = \vec{J} \, dV$

$$= \frac{\mu_0 eN}{8\pi^2} \frac{\hat{r}' \times (\vec{r} - \vec{r}')}{r'^2 |\vec{r} - \vec{r}'|^3} dV'$$

$$= \frac{\mu_0 eN}{8\pi^2} \frac{\hat{r}' \times \vec{r}}{r'^2 |\vec{r} - \vec{r}'|^3} dV'$$

助教打错了!

对于指定 r, θ:

$r'^2 |\vec{r} - \vec{r}'|^3 = \text{Const.}$ $\quad \hat{r}' \times \vec{r} = r \hat{z}$ 积分一圈 $= 0$

$\rightarrow dB(r') = 0$

$B = 0$

$B(r) = 0 \rightarrow B = 0$ (对称性)

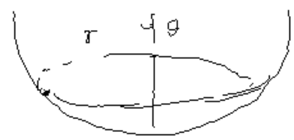
↳ $\nabla \times B = 0 \neq j(r)$? 原因:有位移电流 j_d = 零

例 5.2.8

$$j(r) = \frac{I_0}{2\pi r^2}$$

地面以上 $B(r) = \frac{\mu_0 I}{2\pi r}$

地面以下



$$S_{球} = 2\pi r H$$

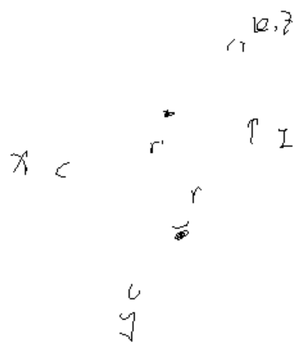
$$= 2\pi r^2 (1-\cos\theta)$$

$$I = j \cdot S_{球} = I_0 (1-\cos\theta)$$

$$B(r) = \frac{1}{2\pi r \sin\theta} \mu_0 I$$

$$= \frac{\mu_0 I_0}{2\pi} \cdot \frac{1-\cos\theta}{r\sin\theta}$$

例 5.2.9.



$$dB(x,z) = \frac{\mu_0 I}{4\pi} \frac{d\vec{l} \times \vec{r'}}{|r'|^3}$$

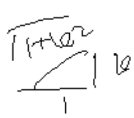
$d\vec{l} = (dz)\,\hat{z}$ $\quad I = k\,dx$ $\quad \vec{r'} = (-x, r, -z)$

$$= \frac{\mu_0 k}{4\pi} dx\,dz \frac{-x\hat{y} - r\hat{x}}{(x^2+z^2+r^2)^{3/2}}$$

$$= -\frac{\mu_0 k}{4\pi} \frac{r\hat{x} + x\hat{y}}{(x^2+z^2+r^2)^{3/2}} dx\,dz$$

$k = \frac{x}{a}$

$$\int \frac{dx}{(a^2+x^2)^{3/2}} = \frac{1}{a^2}\int \frac{dk}{(1+k^2)^{3/2}}$$



$k = \tan t$

$$= \frac{1}{a^2}\int \frac{1}{\frac{1}{\cos^3 t}} \frac{dt}{\cos^2 t} = \frac{1}{a^2}\sin t$$

$$= \frac{1}{a^2}\frac{k}{1+k^2} = \frac{1}{a^2}\frac{x}{(a^2+x^2)^{1/2}}$$

$$B = -\frac{\mu_0 k}{4\pi}\int_{-\infty}^{+\infty} dx \int_{-\infty}^{+\infty} \frac{r\hat{x} + x\hat{y}}{(r^2+x^2+z^2)^{3/2}} dz$$

$$= -\frac{\mu_0 k}{4\pi}\int_{-\infty}^{+\infty} dx\, \frac{r\hat{x}+x\hat{y}}{r^2+x^2}\, \frac{z}{(r^2+x^2+z^2)^{1/2}}\Bigg|_{z=-\infty}^{z=+\infty}$$

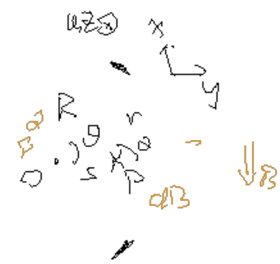
$$= -\frac{\mu_0 k}{2\pi}\int_{-\infty}^{+\infty}\frac{r\hat{x}+x\hat{y}}{r^2+x^2}\,dx \qquad \int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a}$$

$$= -\frac{\mu_0 k}{2\pi}\, r\cdot\frac{1}{r}\arctan\left(\frac{x}{r}\right)\Bigg|_{x=-\infty}^{x=+\infty}\hat{x}$$

$$= -\frac{\mu_0 k}{2}\hat{x}$$

例 5.1.41.

$$\lambda = \frac{I}{2\pi R}$$



$dB_y = 0$ (由对称性)

$$dB_x = 2\,dB\cos\alpha$$

$$= 2\cdot\frac{\mu_0\, dI}{2\pi r}\cos\alpha = \frac{\mu_0\, dI}{\pi r^2}(R\cos\theta - s)$$

$$= \frac{\mu_0 I}{2\pi^2}\frac{(R\cos\theta - s)}{r^2}d\theta = \frac{\mu_0 I}{2\pi^2}\frac{(R\cos\theta - s)\,d\theta}{R^2+s^2-2Rs\cos\theta}$$

$$I_1 = -\int_0^{\pi}\frac{s\,d\theta}{R^2+s^2-2Rs\cos\theta} = -\int_0^{\pi/2}\frac{2s\,dt}{(R^2+s^2)(\sin^2t+\cos^2t)-2Rs(\cos^2t-\sin^2t)}$$

$$= -\int_0^{\pi/2}\frac{2s\,dt}{(R+s)^2\sin^2t+(R-s)^2\cos^2t} = -\int_0^{\pi/2}\frac{2s\,d(\tan t)}{(R+s)^2\tan^2t+(R-s)^2}$$

$$= \frac{-2s}{(R+s)^2}\int_0^{\pi/2}\frac{d(\tan t)}{\left(\frac{R-s}{R+s}\right)^2+\tan^2t} = -\frac{2s}{(R+s)^2}\cdot\left(\frac{R+s}{R-s}\right)\arctan\left(\frac{\tan t}{\frac{R-s}{R+s}}\right)\Bigg|_{t=0}^{t=\frac{\pi}{2}}$$

$$= -\frac{\pi s}{R^2-s^2}$$

$$I_2 = \int_0^{\pi}\frac{R\cos\theta\,d\theta}{R^2+s^2-2Rs\cos\theta} = \int_0^{\pi}\left[\frac{-\frac{R^2+s^2}{2s}+R\cos\theta}{R^2+s^2-2Rs\cos\theta}+\frac{R^2+s^2}{2s}\,\frac{1}{R^2+s^2-2Rs\cos\theta}\right]d\theta$$

$$= -\frac{1}{2s}\cdot\pi + \frac{R^2+s^2}{2s}\,\frac{\pi}{R^2-s^2}$$

$$B_x = \int_0^{\pi} dB_x$$

$$= -\frac{\pi}{2s} + \frac{\pi}{R^2-s^2}\left(\frac{R^2+s^2}{2s} - s\right) = 0 \qquad (s<R)$$

$s = R$ 时:

$$B_x = \frac{\mu_0 I}{4\pi^2 R}\int_0^{\pi}\frac{\cos\theta-1}{1-\cos\theta}\,d\theta = -\frac{\mu_0 I}{4\pi R}$$

$S>R$ 时：

$$B_x = \frac{\mu_0 I}{2\pi^2}\left(-\frac{\pi}{2S} + \left(\frac{R^2+S^2}{2S} - S\right)\cdot\frac{\pi}{S^2-R^2}\right) = \frac{\mu_0 I}{2\pi^2}\cdot -\frac{\pi}{S}$$

$$= -\frac{\mu_0 I}{2\pi S}$$

$$\Rightarrow B|_{S=R} = \frac{1}{2}(B_{S\to R^-} + B_{S\to R^+})$$

例 6.2.14

$$m = IS = Ia^2$$

$$M = \vec{m}\times\vec{B} = Ia^2 B\sin\theta$$

$$J\ddot{\theta} = -M \approx -Ia^2 B\sin\theta$$

$$T = 2\pi\sqrt{\frac{J}{Ia^2B}}$$

例 6.2.35

$$B = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4\pi r} = \frac{3\mu_0 I}{4\pi r}$$

$$dF = B\cdot\frac{I}{2\pi r}Rd\theta\cdot dl$$

$$\frac{dF}{dS} = \frac{dF}{Rd\theta dl} = \frac{3\mu_0 I^2}{8\pi^2 r^2}$$

例 6.2.37

证明两个闭合电流之间作用力满足牛顿第三定律

$$B_{12} = \oint_{l_1}\frac{\mu_0 I_1}{4\pi}\frac{d\vec{l_1}\times\vec{r_{12}}}{|\vec{r_{12}}|^3} \qquad I dl$$

$$F_{12} = \oint_{l_2} I_2 d\vec{l_2}\times\vec{B_{12}} = \frac{\mu_0 I_1 I_2}{4\pi}\oint_{l_2} dl_2\times\left(\oint_{l_1}\frac{d\vec{l_1}\times\vec{r_{12}}}{|\vec{r_{12}}|^3}\right)$$

$$= \oint_{l_2}\oint_{l_1}\frac{d\vec{l_2}\times(d\vec{l_1}\times\vec{r_{12}})}{|\vec{r_{12}}|^3}$$

$$= \oint_{l_2}\oint_{l_1}\frac{1}{|\vec{r_{12}}|^3}\left[(\vec{r_{12}}\cdot d\vec{l_2})d\vec{l_1} - (d\vec{l_1}\cdot d\vec{l_2})\vec{r_{12}}\right]$$

考察 $\oint_{l_2}\oint_{l_1}\frac{1}{|\vec{r_{12}}|^3}(\vec{r_{12}}\cdot d\vec{l_2})\,d\vec{l_1}$

$$= \oint_{l_2}\oint_{l_1}(-\nabla\tfrac{1}{r}\cdot d\vec{l_2})\,d\vec{l_1} = \oint_{l_1}\oint_{l_2}(-\nabla\tfrac{1}{r}\cdot d\vec{l_2})\,d\vec{l_1}$$

其中 $\oint_{l_2} (\nabla \frac{1}{r_{12}}) \cdot d\vec{l_2}$

$= \oint_{l_2} \nabla \frac{1}{r_{12}} \cdot d\vec{r_{12}} = \oint_{l_2} d(\frac{1}{r_{12}}) = 0$

$\Rightarrow \quad F_{12} = -\oint_{l_2}\oint_{l_1} \frac{1}{|r_{12}|^3} (d\vec{l_1} \cdot d\vec{l_2}) \vec{r_{12}}$

$r_{12} = -r_{21}$

$= -F_{21} = -\oint_{l_1}\oint_{l_2} \frac{1}{|r_{21}|^3} (d\vec{l_2} \cdot d\vec{l_1}) \vec{r_{21}}$